

INFORMATION-SECURITY-GRUYERE

Table of Contents

1. Network Design	2
2. Wireless Configuration	2
Explanation:	3
3. Security Configuration	3
Explanation	3
4. Connectivity Testing and Evidence	4
5. Threat Awareness	5
Identified Threats	5
How Security Measures Reduce These Threats	5
Conclusion	5

Design and Analysis of a Secure Wireless Network including Smart Surveillance System

1. Network Design

A secure wireless network was designed using Cisco Packet Tracer to simulate a small office surveillance system.

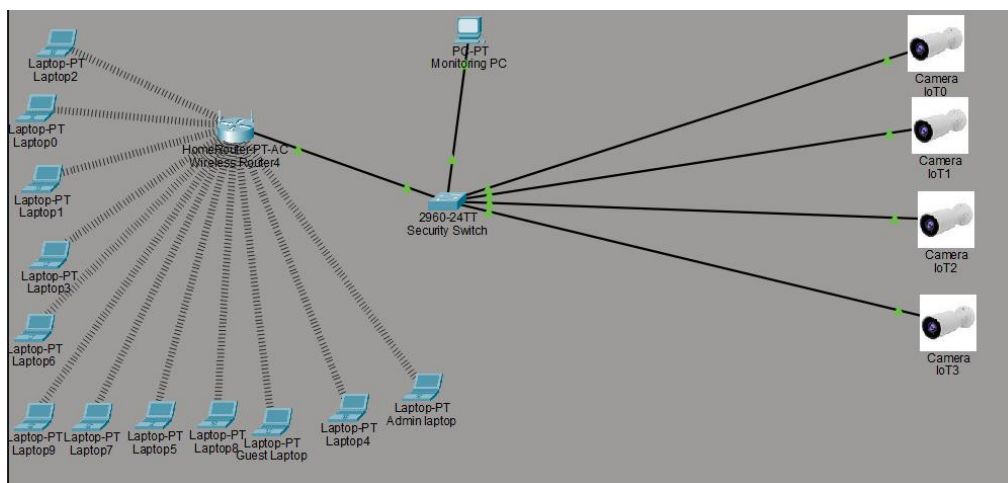
The network consists of one wireless router acting as the central device, which manages communication between all connected devices. A switch labeled **Security Switch** is connected to the router to support wired connections.

Four security cameras were created as IoT devices and connected to the Security Switch using wired Ethernet connections. These cameras are used for monitoring purposes and transmit data to the network.

A monitoring PC is connected to the switch to observe and manage surveillance data. Additionally, ten laptops are connected wirelessly to the network using DHCP, which automatically assigns IP addresses.

An administrator laptop is also connected wirelessly and is used to access the router's GUI for configuration and management purposes.

All devices in the topology are clearly labeled for better understanding and organization.



Network Topology

2. Wireless Configuration

The wireless network was configured using basic settings to ensure proper communication and performance.

The SSID (Service Set Identifier) of the network was set to **Office_Surveillance_Network**, which allows users to identify and connect to the correct network.

The network operates on the **5 GHz frequency band (5GHz-1)**, which provides higher speed and less interference compared to 2.4 GHz.

A guest network was also configured within the same router. The SSID for the guest network was set to **Guest_Network**, with WPA2 security enabled. Guest users are restricted from accessing the main network by disabling local network access.

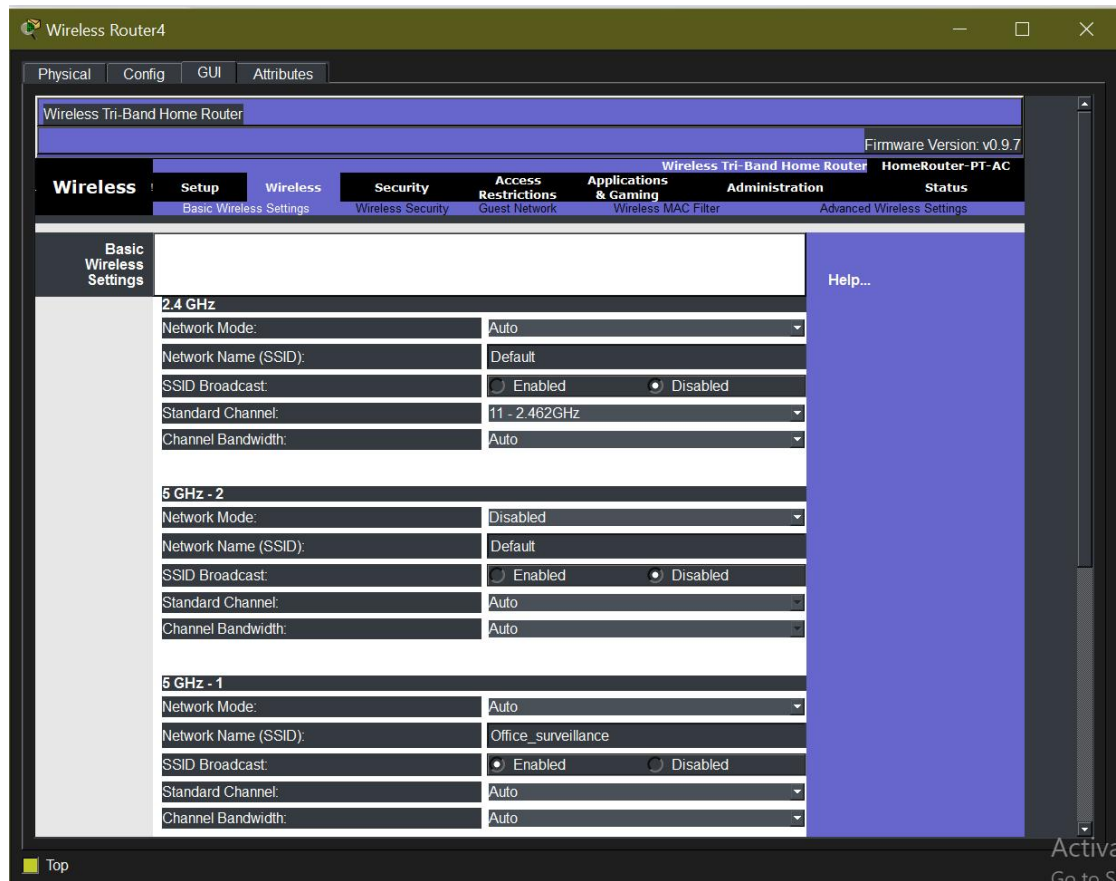
Explanation:

Q1: Why is SSID important?

SSID is important because it uniquely identifies a wireless network and helps users connect to the correct network.

Q2: What is the difference between 2.4 GHz and 5 GHz wireless frequencies?

The 2.4 GHz band offers wider coverage but slower speeds and higher interference, while the 5 GHz band provides faster speeds and reduced interference, making it suitable for office environments.



Basic Wireless Settings

3. Security Configuration

To secure the wireless network, WPA2 encryption was applied along with a strong password.

The security mode was set to **WPA2 Personal with AES encryption**, and a strong password was configured to prevent unauthorized access.

Explanation

Q1: Why is encryption important?

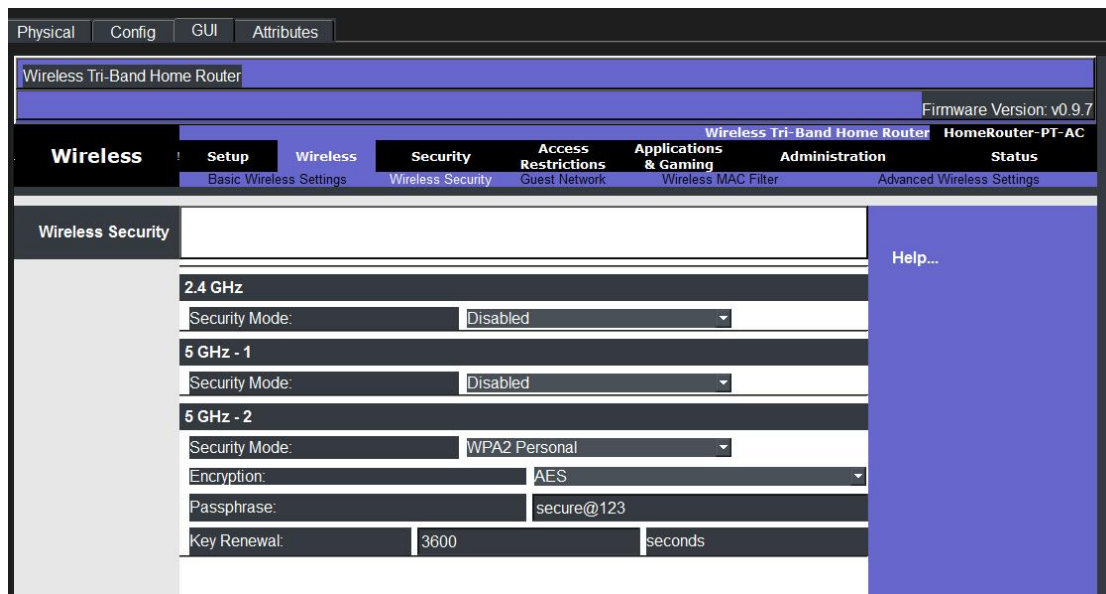
Encryption is important because it protects data by converting it into a secure format, preventing unauthorized users from accessing it.

Q2: What are the risks of an open wireless network?

An open wireless network is vulnerable to risks such as unauthorized access, data theft, and misuse of network resources.

Q3: How can attackers exploit weak security?

Attackers can exploit weak security through password cracking, eavesdropping, and man-in-the-middle attacks, allowing them to gain access to sensitive information.



Wireless Security

4. Connectivity Testing and Evidence

Connectivity between devices was verified using the **ping command**.

The monitoring PC successfully communicated with all security cameras, confirming that the network is functioning properly.

Successful ping replies indicate that devices are correctly connected and able to exchange data within the network.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.125

Pinging 192.168.0.125 with 32 bytes of data:

Reply from 192.168.0.125: bytes=32 time<1ms TTL=255
Reply from 192.168.0.125: bytes=32 time<1ms TTL=255
Reply from 192.168.0.125: bytes=32 time<1ms TTL=255
Reply from 192.168.0.125: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.125:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Pinging results

Access Restrictions

Wireless Tri-Band Home Router HomeRouter-PT-AC

Setup Wireless Security **Access Restrictions** Applications & Gaming Administration Status

Internet Access Policy

Internet Setup

Access Policy: 1() Delete This Entry Summary

Enter Policy Name

Status: ☒ Enabled ☐ Disabled

Applied PCs

Edit List (This Policy applies only to PCs on the List.)

Access Restriction

☐ Always

☒ Never

☐ Specific Time

Schedule

Days: ☒ EveryDay ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri ☐ Sat ☐ Sun

Times: 00:00 to 00:00

Website Blocking by URL Address

URL 1 URL 3

URL 2 URL 4

Website Blocking by Keyword

Keyword 1 Keyword 3

Keyword 2 Keyword 4

Blocked Applications

Note: only three applications can be blocked per policy.

Applications

Ping(0-0)

HTTP(80-80)

HTTPS(443-443)

FTP(21-21)

POP3(110-110)

IMAP(143-143)

SMTP(25-25)

NNTP(119-119)

Telnet(23-23)

SNMP(161-161)

TFTP(69-69)

Blocked List

Help...

Router GUI Access

5. Threat Awareness

Several potential threats exist in a wireless network environment.

Identified Threats

1. Unauthorized Access

Attackers may attempt to connect to the network without permission.

2. Data Interception

Sensitive information transmitted over the network may be captured by attackers.

3. Denial of Service (DoS) Attack

Attackers may overload the network and disrupt its functionality.

How Security Measures Reduce These Threats

These threats are reduced by implementing security measures such as WPA2 encryption, strong passwords, and guest network isolation.

- WPA2 encryption protects data from being intercepted
- Strong passwords prevent unauthorized access
- Guest network isolation ensures that external users cannot access internal devices

Conclusion

This assignment demonstrated the design and implementation of a secure wireless network for a smart surveillance system. Proper configuration of wireless settings, security measures, and connectivity testing ensured that the network operates efficiently and securely.